

EMBEDDING AGENTIC TRACES FOR EFFICIENT EVALUATION

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ABSTRACT

Evaluating agentic systems is expensive: a single leaderboard entry requires hundreds of long-horizon task executions, and modern systems are sensitive to configuration details that multiply the number of variants needing evaluation. We study the problem of predicting a new system’s full-benchmark score from a small number of probe runs, using publicly cached runs of other systems as references. Each probe produces an *agentic trace* that carries far more information than its pass/fail grade, but off-the-shelf semantic embeddings of traces are dominated by evaluation-irrelevant confounders — harness, model family, and exact configuration — rather than by the content of the work performed. We introduce QUBRIC, a query-specific-rubric transformation that converts off-the-shelf embedders into content-faithful, confounder-invariant trace representations, and combine the resulting trace geometry with item response theory in a simple blend. On SWE-bench Verified (107 systems, 100-task panel), the blend predicts full-benchmark resolve rates within 0.066 MAE from a single probe and 0.032 from twenty, is never worse than the strongest correctness-only estimator, and is significantly better exactly where graded evidence is scarce. The reference geometry is constructed from public archives for approximately \$26.

1 INTRODUCTION

The dominant paradigm of generative model use is increasingly agentic: systems that plan, invoke tools, observe results, and iterate over long horizons, rather than emit a single response. Evaluation is a necessary part of understanding an agent’s capability and utility, and as agents become more capable and are given more trust, the tasks on which they must be evaluated become correspondingly complex. Recent agentic evaluations — SWE-bench and its variants (Jimenez et al., 2024; OpenAI, 2024), Terminal-Bench (Terminal-Bench Team, 2026), ARC-AGI (Chollet, 2019), and multi-benchmark agent suites (Kapoor et al., 2026) — require up to hundreds of thousands of tokens per task¹ and evaluate each system on roughly one hundred to two thousand tasks. At published per-task costs of \$0.13–\$4 (aggregate \$1.84/rollout across 21,730 rollouts in Kapoor et al., 2026), this translates to roughly \$100–\$2,000 per full evaluation of a single system, and tens of thousands of dollars for multi-benchmark sweeps. Given the sensitivity of these systems to changes in system prompts, retrieval databases, scaffold parameters, and model versions — each of which creates a new variant worth measuring — these costs are untenable and result in systematic *under-evaluation*: most checkpoints, ablations, and configurations are simply never scored (Kapoor et al., 2024).

In other domains of machine learning evaluation, practitioners have developed efficient techniques to reduce this resource burden: benchmark compression and anchor-item selection (Vivek et al., 2024; Maia Polo et al., 2024), item response theory and adaptive testing (Kipnis et al., 2025; Truong et al., 2025), and performance prediction from observational records (Ruan et al., 2024; Zhuang et al., 2025). One promising approach uses off-the-shelf semantic embedding models to construct low-dimensional representations of the systems themselves and casts benchmark prediction as an inference problem in the low-dimensional space (Duderstadt et al., 2023; Helm et al., 2025; 2026): it requires no labeled training data, no white-box access, and exploits response caches that eval-

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¹In the SWE-bench Verified leaderboard archive we measure a median of ~40K tokens per trace, with a heavy tail exceeding 700K tokens.

uation ecosystems already produce. Unfortunately, off-the-shelf embedding functions do not yet exist for agentic traces; nor are semantic embedding functions, without modification, appropriate for embedding traces for the purpose of efficient evaluation — we show that they are dominated by evaluation-irrelevant *confounders* (harness, model family, exact configuration) rather than by the content of the work performed. We introduce QUBRIC (*query-specific rubrics*) as a way to transform these models into a useful tool for efficient system evaluation.

In this paper we make four contributions. First, we show that off-the-shelf semantic embedding functions are not appropriate for efficient evaluation of agentic systems: their trace representations recover harness, model family, and even exact configuration at far above chance while remaining near chance on outcome, and — diagnostically — their prediction error is *flat* in the number of probe traces. Second, we show that extracting and embedding content from a trace according to a query-specific rubric mitigates these issues: QUBRIC collapses the confounder axes to approximately chance while retaining task and behavioral fidelity, robustly across seven embedding models. Third, we show that QUBRIC representations improve query-efficient evaluation: blended with item-response-theoretic estimators, they are never worse than the strongest correctness-only method at any probe budget and are significantly better where graded evidence is scarce, predicting full-benchmark resolve rates within 0.066 MAE from a single probe run. Fourth, we release the reference geometry, code, and judged-trace corpus — constructed from public archives for approximately \$26 — as infrastructure for trace-based evaluation research.

1.1 PROBLEM STATEMENT

We adapt the query-efficient evaluation framework of Helm et al. (2026) to the agentic setting. Let $\mathcal{T} = \{t_1, \dots, t_N\}$ be a benchmark of N tasks and let an agentic system A be a (stochastic) map from tasks to *traces*: executing A on task t yields a trace $\tau = A(t)$, a long, semi-structured log of the system’s actions, observations, and outputs, together with a grade $g(\tau) \in \{0, 1\}$ assigned by the benchmark harness. The quantity of interest is the system’s benchmark score

$$y(A) = \frac{1}{N} \sum_{i=1}^N \mathbb{E}[g(A(t_i))],$$

estimated exactly only by executing and grading A on all N tasks.

We are given a *reference cache*: systems $A_1, \dots, A_M$ that have already been evaluated in full, so that for each reference j we observe its traces $\{\tau_{ij}\}_{i=1}^N$, its grades, and hence $y(A_j)$. Such caches exist publicly at scale.² A *new* system A_0 is executed on only a probe subset $S \subset \mathcal{T}$ with $m = |S| \ll N$, yielding probe traces $\{\tau_{i0}\}_{i \in S}$ and (optionally) their grades. The goal is an estimator $\hat{y}(A_0)$ minimizing $\mathbb{E}|\hat{y}(A_0) - y(A_0)|$ at a given probe budget m , where the estimator may use the probe traces, the probe grades, and the full reference cache.

Two aspects distinguish the agentic setting from the short-response setting of Helm et al. (2026). First, the observation associated with each probe is not a short answer but a trace of 10^4 – 10^6 tokens in one of many mutually incompatible harness formats; whether and how this observation can be embedded usefully is the central question of this paper. Second, the reference population is confounded: references sharing the target’s underlying LLM or harness permit trivially accurate prediction by score-copying, so we evaluate under a *leave-one-LLM-out* protocol — references sharing the target’s underlying LLM are excluded — and verify robustness under *leave-one-harness-out*. The probe subset S may be fixed, drawn at random, or chosen per-target by adaptive item selection; we report all three regimes.

2 RELATED WORK

Agents, traces, and evaluations. Agentic benchmarks pair long-horizon tasks with automated harnesses: SWE-bench and its verified subset (Jimenez et al., 2024; OpenAI, 2024) for repository-level software issues, executed by scaffolds such as SWE-agent (Yang et al., 2024) and OpenHands (Wang et al., 2025); Terminal-Bench (Terminal-Bench Team, 2026) for command-line work; and

²The SWE-bench Verified leaderboard archive contains trajectories and outcomes for over one hundred systems; we estimate it embodies \$50K–100K of agent compute contributed by submitters.

suites aggregating many such settings (Kapoor et al., 2026). The traces these systems emit have recently become objects of study in their own right: symbolic action-alphabet analyses of leaderboard trajectories (Mehtiyev & Assunção, 2026), canonical action taxonomies that document per-scaffold format fragmentation (Shu et al., 2026), failure taxonomies (Cemri et al., 2025), trajectory-level judging (Zhuge et al., 2025), and trajectory compression (Xiao et al., 2025). This literature produces taxonomies, statistics, and reviews rather than reusable vector representations, and none of it addresses the confounder dominance that its own attribution results imply: coding-agent trajectories are attributable to scaffold and model at 85.7% against an 11% baseline (Oderinwale, 2026), and model-level fingerprints survive paraphrase and summarization (Sun et al., 2025; Wu et al., 2026). Benchmark-integrity findings — memorization inflating resolve rates (Liang et al., 2025) and mislabeled gold patches (Yu et al., 2025) — further motivate representations of *how* systems work rather than reliance on the pass/fail bit alone.

Efficient evaluations. Efficient evaluation methods exploit the approximate low-rank structure of the model $\times$ item success matrix: anchor items and benchmark compression (Vivek et al., 2024; Maia Polo et al., 2024; Kipnis et al., 2025), amortized and adaptive item response theory (Truong et al., 2025), and observational performance prediction (Ruan et al., 2024; Zhuang et al., 2025). These consume only correctness patterns — a few bits per probe — which imposes a documented resolution floor at extreme budgets (Yauney et al., 2026). Agent-specific efficiency work is nascent: subset selection collapses under scaffold shift (Ndzomga, 2026), and psychometric models extended with task features still consume correctness rather than trace content (Ge et al., 2026). Closest to this work is the data kernel perspective space lineage (Duderstadt et al., 2023; Helm et al., 2025; Acharyya et al., 2024), and in particular query-efficient evaluation from cached responses (Helm et al., 2026), which predicts a new model’s score by comparing few probe responses against cached reference coverage. We generalize that framework to the agentic setting, where each probe observation is a trace rather than a short response, and where the reference population’s confounder structure requires an invariant representation and a conservative reference protocol.

Embeddings of non-standard objects. General-purpose text embedders (Reimers & Gurevych, 2019) fix one notion of similarity at training time; no single embedding can represent all relevance relations (Weller et al., 2025), and instruction-conditioned variants (Su et al., 2023) honor only coarse, trained-in task tags. Conditional similarity (Deshpande et al., 2023) formalizes similarity-under-a-criterion at sentence scale, and a family of QA-mediated representations embeds LLM-generated text in place of the original object: answers to an instruction (Peng et al., 2024), answer vectors to question batteries (Benara et al., 2024), and description-based similarity (Ravfogel et al., 2024). Embeddings of interaction data exist for short dialogues (Liu et al., 2022) and RL state-action trajectories (Ge et al., 2025) — the latter observing that trajectory embeddings encode agent “style,” precisely the confounder our setting must remove. QUBRIC sits at the unclaimed intersection: inference-time, training-free, rubric-specified re-representation of very long, heterogeneous logs, with an explicit invariance mechanism (per-task consensus centering) and an evaluation protocol that measures — rather than assumes — confounder invariance.

3 QUBRIC

4 EXPERIMENTS

5 DISCUSSION

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